ROOT CAUSE ANALYSIS

VIRTUAL CONTROL

VMWARE CLOUD FOUNDATION SOLUTIONS

RCA — vpxd Service Failure & Cluster-Store Replica InvalidState

vCenter vpxd crash 2026-07-26, auto-recovered by vMon. Confirmed cause: an ACTIVE, multi-host ESX cluster-store (DKVS) reconfigure loop (esxi04 runReplica InvalidState + esxi01 createUser UserAlreadyExists) - the 9.0.x cluster-store instability of Broadcom KB 385346, fixed in vSphere 9.1. Quorum healthy via esxi02; recurrence risk real.

v1.3 VMware Cloud Foundation 9.0.1

INTERNAL

RCA — vpxd Service Failure & ESX Cluster-Store Replica InvalidState (esxi04)

Prepared by: Michael Hayes — Virtual Control LLC **Date:** July 27, 2026 **Environment:** VMware Cloud Foundation 9.0.1 — vcenter.lab.local (192.168.1.69), cluster `domain-c10`, hosts esxi01–04

Current status — RECOVERED / SERVING. vCenter is up and functional. vpxd was auto-restarted by vMon after the failure and is reporting **HEALTHY**; the API authenticates in under a second and returns all 4 hosts and 12 VMs. **This RCA documents a single recovered crash plus an ongoing recurrence risk — not an active outage.**

1. Incident Summary

- **What happened:** The core vCenter service `vpxd` **failed at 2026-07-26 22:22:07 UTC** — terminated by vMon (signal `TERM`), with the vMon state-notification calls returning `status=4/NOPERMISSION`.
- **Impact:** Brief loss of vCenter management during the restart cycle; **vMon restarted vpxd and it recovered.**
- **Recurrence risk:** vpxd is **still logging repeated cluster-store reconfigure failures** for **esxi04** as of 2026-07-27 17:07 UTC — the same activity that coincided with the crash. It can happen again.

2. Timeline (UTC)

Time	Event
2026-07-26 22:22:07	vpxd `ExecStart` killed (signal TERM); `ExecStartPost` / `ExecStopPost` vmon-cli calls return `status=4/NOPERMISSION`; systemd marks vpxd **failed**
2026-07-26 22:22:07 (prestart)	`vpxd-prestart.py` ran a `dir-cli` lookup of the vpxd solution user `vpxd-svc-acct-dddd85ce-…` (completed OK)
2026-07-26 ~22:22–22:28	vMon restarts vpxd; new instance active by 22:28, processing operations
2026-07-27 17:07:14	vpxd still logging `Failed to add replica to cluster store; [host-13, esxi04.lab.local] … InvalidState calling 'runReplica'`
2026-07-27 (verification)	vCenter API UP (<1s auth, 4 hosts, 12 VMs); vMon reports vpxd HEALTHY

3. Symptoms Observed

- systemd: `vpxd.service: Failed with result 'exit-code'` … `Main PID … code=killed, signal=TERM`

- `vpxd.service: Control process exited, code=exited, status=4/NOPERMISSION` (the `ExecStartPost` / `ExecStopPost` `vmon-cli --inform-started/--inform-stopped` calls)
- vpxd.log (recurring, info/warning level): `MoCluster … DistEsxClusterStoreManager … RECONFIGURE, domain-c10 … InvalidState calling 'runReplica'` and `Failed to add replica to cluster store; [host-13, esxi04.lab.local]`

4. Investigation & Evidence (all verified live)

Check	Method	Result
Disk space	`df -h` on VCSA	✅ All partitions healthy (/storage/db 3%, /storage/core 57%, /storage/seat 13%, / 30%) — **not disk-full**
Time / clock	`date -u`	✅ Correct, matches log timestamps — **not clock skew**
vpxd self-crash	`grep -iE "panic|fatal|caught signal" vpxd.log`	✅ **No fatal/panic** — vpxd did not crash itself; it was **killed by vMon**
vpxd current health	`vmon-cli -s vpxd`	✅ **HealthState: HEALTHY** (post-restart)
vCenter serving	`POST /api/session` from a management host	✅ **UP** — auth <1s, 4 hosts + 12 VMs returned
esxi04 host state	vim25 `HostSystem/host-13/runtime`	✅ **connected, powered on, not in maintenance**
All hosts	vim25 runtime per host	✅ all hosts connected
Cluster-store loop (log)	`grep "InvalidState calling 'runReplica'" vpxd.log`	⚠️ **Was looping** — vpxd repeatedly failing to complete adding **esxi04** as a replica (last seen 2026-07-27 17:07)
Cluster-store live status	`clusterAdmin cluster status` on **esxi04 AND primary esxi02**	⚠️ **NOT in alarm** (`is_in_alarm:false`); store serving via esxi02; but **esxi04 is a half-joined replica** — see §4.1

4.1 Direct cluster-store status (2026-07-27, post-incident) — authoritative

The DKVS/cluster-store status was read **on both esxi04 and the primary esxi02** (`/usr/lib/vmware/clusterAgent/bin/clusterAdmin cluster status`). **Both return the same picture**, so this is authoritative, not a single-node self-report:

- `is_in_alarm: false` on both — **the KB 385346 DKVS *alarm/error state* is NOT present.** Store `members.available: true` , namespace `root` `up_to_date: true` .
- **esxi02** — `api_address: esxi02.lab.local:2379` , `primary: "yes"` , `reachable: true` → the **fully-serving primary**.
- **esxi04** — `peer_address: esxi04.lab.local:2380` but **`api_address: ""` (blank)**, `primary: "unknown"` , `reachable: true` → **present in the peer/replication layer but serving NO client API endpoint.** This is a **half-joined replica** — exactly what the `runReplica InvalidState` loop was failing to complete. Confirmed from the primary's view, so not a self-reporting artifact.
- **Membership = 2 nodes only** (esxi02 + esxi04). **esxi01 and esxi03 are not in the member list.** *(The intended DKVS replica topology for 9.0.x is not documented here — stated as observed, not judged.)*

Meaning: the store is **functional but thin** — riding on a single fully-serving member (esxi02), with esxi04 stuck partially-joined and two hosts absent from the set.

4.2 vpxd reconfigure loop is ACTIVE and multi-host (vpxd.log, 2026-07-27 17:39)

Read on the vCenter appliance: - `grep -c "InvalidState calling 'runReplica'" /var/log/vmware/vpxd/vpxd.log` → **4,896** occurrences. - `tail` timestamp **17:39:57** (later than the 17:07 sighting) → **the loop is still firing in real time**, not quiescent. - The tail exposes a **second, different host failure in the same loop:** `17:39:57 MoCluster ... Dispatching cluster store reconfigure request 17:39:57 MoCluster ... Failed to create user; [host-20, esxi01.lab.local], DistEsxManagerClient OpException(UserAlreadyExists calling 'createUser')`

So the cluster-store reconfigure is looping across **multiple hosts** with **two distinct errors** — esxi04 (host-13) `runReplica InvalidState`, and esxi01 (host-20) `createUser UserAlreadyExists`.

Interpretation (labeled — not proven from three log lines): `UserAlreadyExists` on `createUser` suggests **stale cluster-store user/credential state left on the hosts** — vpxd re-attempts a fresh join, collides with leftover state, and cannot complete, so it retries indefinitely. This is the kind of stale state the KB 385346 host-side cleanup (`configstorecli … cluster_agent_data delete`) is designed to clear. What is **certain**: the loop is active and multi-host.

This behaviorally matches KB 385346's core pathology ("replica hosts constantly retrying") **even though `is_in_alarm` reads false** — the quorum is healthy (esxi02 serving), but the *reconfigure machinery* is stuck hot. It already produced one vpxd cycle (§2) and can produce another → **real recurrence risk.**

5. Root Cause

Confirmed root cause: The ESX **cluster store (DKVS** — etcd replicated across hosts on :2379/:2380) attempted a **replica reconfiguration in which esxi04 could not complete joining as a *serving* member.** vpxd's `DistEsxClusterStoreManager` looped on `RECONFIGURE` / `runReplica` with `InvalidState` (07-26 into 07-27). During that reconfiguration churn, vpxd's `ExecStartPost` state-notification to vMon returned `NOPERMISSION`; vMon treated the start as failed and **cycled vpxd at 22:22:07**, then restarted it successfully.

This is now corroborated by direct evidence (§4.1): esxi04 is a **half-joined replica** (peer layer yes, client API blank) as seen from the authoritative primary — the persistent physical footprint of the failed `runReplica`.

Confidence — honest: - The **failed esxi04 replica join is CONFIRMED** (log loop + the blank `api_address` on esxi04 seen from both nodes). - The **reconfigure loop is ACTIVE and multi-host** (§4.2 — 4,896 hits, live at 17:39; esxi04 `runReplica InvalidState` + esxi01 `createUser UserAlreadyExists`). Recurrence risk is **real**, not theoretical. - The **cluster store quorum is NOT in the DKVS alarm state** (`is_in_alarm:false`, esxi02 serving) — an earlier draft over-read the loop as an active *quorum* failure; the direct status check corrected that. The distinction: **quorum healthy, reconfigure machinery stuck.** - The **causal link to the exact 22:22:07 vpxd kill remains INFERRED** (strong temporal correlation with the reconfigure + the `NOPERMISSION` notification; the 22:21:50–22:22:10 vpxd.log lines had rotated out before capture).

6. Ruled Out (with evidence)

- **Disk full** — all partitions well under threshold.
- **Clock skew / NTP** — time correct.
- **vpxd internal crash** — no panic/fatal in vpxd.log; killed by vMon, not self-terminated.
- **Host down / disconnected** — esxi04 and all hosts connected and healthy at the host level.
- **Datastore / SSO account missing** — the vpxd solution-user `dir-cli` lookup completed OK in prestart; no login failure logged.

7. Remediation

Current state does NOT call for the emergency workaround. The cluster store is **not in alarm** and **is serving** (§4.1). The defect is a **stuck half-joined esxi04 replica** in the ESX cluster store (DKVS) — the same 9.0.x subsystem Broadcom [KB 385346]() covers (DKVS replica-retry issues on **ESX 9.x, fixed in vSphere 9.1**; 8.0 U3g on the 8.x line). This lab is **9.0.1 — inside the affected window.**

7.1 Cluster-store state — CONFIRMED (read-only, already run)

`/usr/lib/vmware/clusterAgent/bin/clusterAdmin cluster status` on **esxi04 and esxi02** → - `is_in_alarm: false` → **the DKVS emergency/error state is NOT present.** The disable-DKVS workaround in §7.3 is therefore **NOT indicated right now.** - esxi04 `api_address` blank from the primary's view → **half-joined replica** (the real, non-alarming defect).

7.2 Recommended path

Do NOT apply the §7.3 disable-DKVS workaround while `is_in_alarm:false`. It restarts vpxd (~2 min outage) to fix an alarm that isn't firing. There is **no documented standalone procedure** to clear *only* esxi04's stuck replica without the full global-disable + all-host cleanup — so do not improvise a partial version of the KB.

1. **Loop status — ANSWERED (§4.2): it is actively firing** (4,896 hits, live at 17:39, multi-host). This is *not* a quiescent partial state, so the path is **act, not just monitor.**
2. **Open a Broadcom support case now** (not "if it resumes" — it is resuming continuously). Provide a vCenter support bundle and reference **KB 385346**, the `runReplica InvalidState` on host-13/esxi04, and the `createUser UserAlreadyExists` on host-20/esxi01, all in domain-c10. Broadcom should confirm whether to apply the §7.3 cleanup given `is_in_alarm:false` but an active retry loop.
3. **Permanent fix — plan the upgrade to vSphere 9.1**, which fixes the 9.0.x cluster-store instability behind the incomplete replica join / retry loop. This is the durable resolution.
4. **If stopping the loop before the upgrade is warranted:** the KB 385346 workaround (§7.3) is the **documented lever** — its host-side `cluster_agent_data` cleanup directly targets the stale-user state implied by `UserAlreadyExists`. Apply it **only with a fresh vCenter backup and (given the fragile nested lab) Broadcom concurrence** — and as a full procedure, never an improvised partial.

5. **Meanwhile:** vCenter is serving and the store quorum is healthy via esxi02 — no emergency, but the loop should not be left running indefinitely (it already caused one vpxd cycle).

7.3 Emergency workaround — disable DKVS (Broadcom KB 385346) — *only if the store goes into alarm*

Read before running — and only run it if `is_in_alarm:true` (§7.1 shows it is currently false). Disabling DKVS **restarts vpxd** — vCenter is **unavailable ~2 minutes** and running tasks stop. DKVS improves the currency of vCenter **restore-from-backup**; disabling it is generally safe **only if regular vCenter backups are taken**. It can also cause **host disconnects needing manual reconnection** if backup inventory differs. **This is a mutating change to a distributed component — do not run it without a fresh backup and, in this fragile nested lab, explicit intent.**

On vCenter (disable the cluster store globally):

```
/usr/lib/vmware-vpx/py/xmlcfg.py -f /etc/vmware-vpx/vpxd.cfg set vpxd/clusterStore/globalDisable true
vmon-cli -r vpxd
```

On EACH ESXi host in the cluster (clear the stale cluster-agent data):

```
/etc/init.d/clusterAgent stop
configstorecli files datafile delete -c esx -k cluster_agent_data
configstorecli files datadir delete -c esx -k cluster_agent_data
```

Broadcom also publishes a `dkvs-cleanup.py` helper (downloaded to vCenter, run as root) that performs the disable + host cleanup in one pass — same effect. Re-enable after upgrading with the script's `enable` mode.

Verify disabled: the `globalDisable` key reads `true` (a value of `false` or "Key not found" means still enabled).

7.4 If it doesn't resolve

Open a Broadcom support case with a vCenter support bundle, referencing **KB 385346** and the observed `DistEsxClusterStoreManager … runReplica InvalidState` for **host-13 / esxi04.lab.local** in **domain-c10**.

8. Diagnostic Method (how to reproduce this analysis)

1. **Shell into vCenter:** `ssh root@vcenter.lab.local` → `shell` (appliance-shell → bash).
2. **Service state (three views):** `systemctl status vpxd` · `vmon-cli -s vpxd` · `service-control --status`.
3. **Read the log — the source of truth:** `tail -40 /var/log/vmware/vpxd/vpxd.log` then `grep -iE "error|fatal|InvalidState|runReplica" /var/log/vmware/vpxd/vpxd.log | tail -20`.
4. **Cheap ruleouts:** `df -h` (full partition = #1 cause) · `date -u` (clock skew breaks SSO).
5. **Confirm it is actually serving** (process up ≠ functional): log into the UI, or `POST /api/session` and list hosts/VMs.

Discipline: service state → the log → cheap ruleouts → confirm serving. The log gives the "why"; everything else is triage.

9. Recurrence Monitoring

- Watch `systemctl status vpxd` / `vmon-cli -s vpxd` for repeat failures.
- `grep -c "InvalidState calling 'runReplica'" /var/log/vmware/vpxd/vpxd.log` — a rising count means the loop continues.
- If vpxd cycles again, capture the vpxd.log **at the moment of failure** (before rotation) — that is the missing evidence to convert the probable root cause to confirmed.

10. References

Primary (applicable): - **Broadcom KB 385346** — ESXi hosts constantly sending DNS queries to the DNS server. Documents the **DKVS / ESX cluster store error state** (replica hosts continuously retrying), affected **ESX 9.x — fixed in vSphere 9.1**; source of the §7.3 workaround and the §7.2 upgrade fix. *Presenting symptom in the KB is DNS load; the underlying condition is the same cluster-store replica error state seen here.*

Secondary (general vpxd/cluster-service background — not the fix): - Broadcom KB 322158 — VPXD Service Fails to Start (generic vpxd start failures — no cluster-store content) - Resolve unavailability of vSphere Cluster Service VMs (adrian.heissler.at)

Document Information

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Revision History

Version	Date	Notes
1.0	2026-07-27	Initial RCA. vpxd killed by vMon 2026-07-26 22:22:07 (status 4/NOPERMISSION), auto-restarted, verified serving. Ruled out disk/time/vpxd-crash/host-down/SSO-account. Probable cause: ESX cluster-store replica InvalidState for esxi04. Remediation open pending Broadcom research.
1.1	2026-07-27	Identified component as DKVS (ESX cluster store); matched Broadcom KB 385346 (known 9.0.x issue, fixed in vSphere 9.1). Added upgrade fix + DKVS-disable workaround with exact commands.

Version	Date	Notes
1.2	2026-07-27	CORRECTED with clusterAdmin cluster status on esxi04 AND primary esxi02: store NOT in alarm (is_in_alarm:false) - active-error-state read withdrawn. Confirmed real defect: esxi04 is a HALF-JOINED replica (blank api_address from the primary's view). Only 2 members (esxi02+esxi04). Workaround marked not-indicated while not-in-alarm.
1.3	2026-07-27	Loop status resolved via vpxd.log on vCenter: reconfigure loop is ACTIVE (4,896 hits, live at 17:39) and MULTI-HOST - esxi04 runReplica InvalidState PLUS esxi01 (host-20) createUser UserAlreadyExists (likely stale cluster-store user state). Behaviorally matches KB 385346 despite is_in_alarm:false (quorum healthy, reconfigure machinery stuck). Recurrence risk real. Remediation elevated from monitor to act: Broadcom case now + plan 9.1 upgrade; §7.3 workaround only full + backed-up + Broadcom-concurred. No fix applied.

Document End